DVLM
Assignment 1

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1 Introduction

We solve Problems 1–6 using the assignment notation:

$$q(x_0), \quad q(x_{1:L} \mid x_0), \quad q(x_i \mid x_{i-1}), \quad p_\theta(x_{0:L}), \quad p_\theta(x_{i-1} \mid x_i), \quad p(x_L) = \mathcal{N}(0, I).$$

Also,

$$\alpha_i := 1 - \beta_i, \qquad \bar{\alpha}_i := \prod_{j=1}^{i} \alpha_j, \qquad \bar{\alpha}_0 = 1.$$

2 Problem 1: Forward Process

1.1 Derive the closed-form marginal using recursion

From the assignment, one forward step is

$$x_i = \sqrt{\alpha_i}\, x_{i-1} + \sqrt{1 - \alpha_i}\, z_i, \qquad z_i \sim \mathcal{N}(0, I),$$

with $\alpha_i = 1 - \beta_i$ and $\bar{\alpha}_i = \prod_{j=1}^{i} \alpha_j$.

Recursive idea (simple). Each step does two things:

- multiplies the previous signal by $\sqrt{\alpha_i}$,
- adds fresh Gaussian noise with variance $(1 - \alpha_i)I$.

So after many steps, signal gets multiplied repeatedly, while noise accumulates.

Base case ($i = 1$).

$$x_1 = \sqrt{\alpha_1} x_0 + \sqrt{1 - \alpha_1} z_1 = \sqrt{\bar{\alpha}_1} x_0 + \sqrt{1 - \bar{\alpha}_1}\, \epsilon_1,$$

so the formula is true for $i = 1$.

Inductive step. Assume for $i - 1$:

$$x_{i-1} = \sqrt{\bar{\alpha}_{i-1}} x_0 + \sqrt{1 - \bar{\alpha}_{i-1}}\, \epsilon_{i-1}, \quad \epsilon_{i-1} \sim \mathcal{N}(0, I).$$

Plug into one-step recursion:

$$x_i = \sqrt{\alpha_i}\Big(\sqrt{\bar{\alpha}_{i-1}} x_0 + \sqrt{1 - \bar{\alpha}_{i-1}}\, \epsilon_{i-1}\Big) + \sqrt{1 - \alpha_i}\, z_i.$$

So

$$x_i = \sqrt{\alpha_i \bar{\alpha}_{i-1}}\, x_0 + \underbrace{\sqrt{\alpha_i (1 - \bar{\alpha}_{i-1})}\, \epsilon_{i-1} + \sqrt{1 - \alpha_i}\, z_i}_{\text{sum of independent Gaussians}}.$$

The noise part is Gaussian with variance

$$\alpha_i(1-\bar{\alpha}_{i-1}) + (1-\alpha_i) = 1 - \alpha_i\bar{\alpha}_{i-1} = 1 - \bar{\alpha}_i.$$

Also $\alpha_i\bar{\alpha}_{i-1} = \bar{\alpha}_i$. Therefore

$$x_i = \sqrt{\bar{\alpha}_i}\, x_0 + \sqrt{1-\bar{\alpha}_i}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I),$$

which is exactly Equation (2), i.e.

$$q(x_i \mid x_0) = \mathcal{N}\big(\sqrt{\bar{\alpha}_i}x_0,\, (1-\bar{\alpha}_i)I\big)\,.$$

1.2 SNR monotonicity

$$\mathrm{SNR}(i) = \frac{\bar{\alpha}_i}{1-\bar{\alpha}_i}.$$

Under standard schedules $0 < \beta_i < 1 \Rightarrow 0 < \alpha_i < 1$, so $\bar{\alpha}_i$ decreases with i. Since $a/(1-a)$ is increasing on $(0,1)$, $\mathrm{SNR}(i)$ also decreases with i.

Interpretation: early steps are easier (high SNR), late steps are harder (low SNR).

1.3 If $\bar{\alpha}_i$ decays too quickly, what fails first?

Most immediate failure is **learnability of reverse conditionals** (B). Why: when $\bar{\alpha}_i$ collapses too early, x_i becomes nearly pure noise too soon, so x_i carries weak information about x_0. Then matching

$$q(x_{i-1} \mid x_i, x_0) \approx p_\theta(x_{i-1} \mid x_i)$$

in the diffusion KL term of (11) becomes difficult. This hurts training before sampling stability becomes the main issue.

3 Problem 2: Key Gaussian Posterior

2.1 Derive (5)–(7)

Using

$$q(x_i \mid x_{i-1}) = \mathcal{N}(\sqrt{\alpha_i}x_{i-1}, \beta_i I), \quad q(x_{i-1} \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_{i-1}}x_0, (1-\bar{\alpha}_{i-1})I),$$
$$q(x_{i-1} \mid x_i, x_0) \propto q(x_i \mid x_{i-1})q(x_{i-1} \mid x_0).$$

Complete the square in x_{i-1}:

$$-\frac{1}{2}\left[\frac{\|x_i - \sqrt{\alpha_i}x_{i-1}\|^2}{\beta_i} + \frac{\|x_{i-1} - \sqrt{\bar{\alpha}_{i-1}}x_0\|^2}{1-\bar{\alpha}_{i-1}}\right].$$

Posterior variance:

$$\tilde{\beta}_i = \left(\frac{\alpha_i}{\beta_i} + \frac{1}{1-\bar{\alpha}_{i-1}}\right)^{-1} = \frac{1-\bar{\alpha}_{i-1}}{1-\bar{\alpha}_i}\beta_i.$$

Posterior mean:

$$\tilde{\mu}_i(x_i, x_0) = \frac{\sqrt{\bar{\alpha}_{i-1}}\beta_i}{1-\bar{\alpha}_i}x_0 + \frac{\sqrt{\alpha_i}(1-\bar{\alpha}_{i-1})}{1-\bar{\alpha}_i}x_i.$$

Hence

$$q(x_{i-1} \mid x_i, x_0) = \mathcal{N}(\tilde{\mu}_i, \tilde{\beta}_i I).$$

2.2 Show $\tilde{\beta}_i < \beta_i$ for $i \geq 2$

$$\tilde{\beta}_i = \frac{1-\bar{\alpha}_{i-1}}{1-\bar{\alpha}_i}\beta_i.$$

Since $\bar{\alpha}_i = \alpha_i \bar{\alpha}_{i-1} < \bar{\alpha}_{i-1}$, we have

$$1-\bar{\alpha}_{i-1} < 1-\bar{\alpha}_i \quad \Rightarrow \quad \frac{1-\bar{\alpha}_{i-1}}{1-\bar{\alpha}_i} < 1.$$

Therefore $\tilde{\beta}_i < \beta_i$. Interpretation: conditioning on (x_i, x_0) reduces uncertainty versus one forward step noise.

2.3 Why (5) is a teacher

(5) is the exact posterior of the known forward process. Training minimizes KL between this teacher conditional and the learned reverse conditional $p_\theta(x_{i-1} \mid x_i)$, so it provides ground-truth targets for each step.

4 Problem 3: ELBO Decomposition and Meaning

3.1 Derive (11) from (10), (3)

Start from

$$L_{\text{ELBO}}(x_0;\theta) = \mathbb{E}_{q(x_{1:L}|x_0)}\left[\log q(x_{1:L} \mid x_0) - \log p_\theta(x_{0:L})\right].$$

Using (3),

$$\log p_\theta(x_{0:L}) = \log p(x_L) + \sum_{i=1}^{L} \log p_\theta(x_{i-1} \mid x_i).$$

Use Bayes/Markov identity:

$$q(x_{1:L} \mid x_0) = q(x_L \mid x_0) \prod_{i=2}^{L} q(x_{i-1} \mid x_i, x_0).$$

Then regroup:

$$L_{\text{ELBO}} = \underbrace{\text{KL}(q(x_L \mid x_0)\|p(x_L))}_{L_{\text{prior}}} + \underbrace{\mathbb{E}_{q(x_1|x_0)}[-\log p_\theta(x_0 \mid x_1)]}_{L_{\text{recon}}} + \underbrace{\sum_{i=2}^{L} \mathbb{E}_{q(x_i|x_0)}\left[\text{KL}(q(x_{i-1} \mid x_i, x_0)\|p_\theta(x_{i-1} \mid x_i))\right]}_{L_{\text{diffusion}}}.$$

3.2 Why this is conditional matching

Each KL term in $L_{\text{diffusion}}$ forces

$$p_\theta(x_{i-1} \mid x_i) \approx q(x_{i-1} \mid x_i, x_0)$$

for every i, i.e., reverse-time local transitions are matched step-by-step.

3.3 Precise "L_{prior} is small" statement

Using assignment KL directly:

$$L_{\text{prior}} = \text{KL}(q(x_L \mid x_0)\|p(x_L)), \quad q(x_L \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_L}x_0, (1-\bar{\alpha}_L)I),\ p(x_L) = \mathcal{N}(0, I).$$

Closed form:

$$L_{\text{prior}}(x_0) = \frac{1}{2}\left[\bar{\alpha}_L\|x_0\|_2^2 + d\big((1-\bar{\alpha}_L) - \log(1-\bar{\alpha}_L) - 1\big)\right].$$

If schedule makes $\bar{\alpha}_L \to 0$ (equivalently $\sum_i \beta_i$ sufficiently large), then $L_{\text{prior}} \to 0$ in expectation.

5 Problem 4: ϵ-Prediction, L_{simple}, and What Is Lost

4.1 Derive (8) from (7)

From (2),

$$\hat{x}_0 = \frac{x_i - \sqrt{1-\bar{\alpha}_i}\,\hat{\epsilon}}{\sqrt{\bar{\alpha}_i}}.$$

Substitute $\hat{x}_0$ into posterior mean form (7) and simplify:

$$\mu_\theta(x_i, i) = \frac{1}{\sqrt{\alpha_i}}\left(x_i - \frac{\beta_i}{\sqrt{1-\bar{\alpha}_i}}\epsilon_\theta(x_i, i)\right),$$

which is (8).

4.2 Why diffusion KL becomes weighted MSE in ϵ

For fixed variance $\Sigma_\theta(i) = \tilde{\beta}_i I$, each diffusion KL is

$$\mathrm{KL}\big(\mathcal{N}(\tilde{\mu}_i, \tilde{\beta}_i I) \,\|\, \mathcal{N}(\mu_\theta, \tilde{\beta}_i I)\big) = \frac{1}{2\tilde{\beta}_i}\|\tilde{\mu}_i - \mu_\theta\|_2^2.$$

After substituting $\tilde{\mu}_i$ and μ_θ in ϵ-parameterization:

$$L_{\text{diffusion}} = \sum_{i=2}^{L} \mathbb{E}\big[w_i \|\epsilon - \epsilon_\theta(x_i, i)\|_2^2\big] + \text{const},$$

$$w_i = \frac{\beta_i^2}{2\tilde{\beta}_i \alpha_i (1-\bar{\alpha}_i)}.$$

Dropping w_i gives unweighted L_{simple} in (9).

4.3 What is lost when timestep weights are dropped

You no longer optimize the exact ELBO weighting across timesteps in (11). Common symptom: better sample quality can coexist with worse likelihood/BPD, and timestep-wise error can become imbalanced.

6 Problem 5: Tweedie Identity and Denoising–Score Equivalence

5.1 Prove Tweedie (13)

Define

$$p_\sigma(\tilde{x}) = \int q(x_0)\,\mathcal{N}(\tilde{x}; x_0, \sigma^2 I)\,dx_0.$$

Differentiate:

$$\nabla_{\tilde{x}} p_\sigma(\tilde{x}) = \int q(x_0)\,\mathcal{N}(\tilde{x}; x_0, \sigma^2 I)\frac{x_0 - \tilde{x}}{\sigma^2}\,dx_0.$$

Thus

$$\nabla_{\tilde{x}} \log p_\sigma(\tilde{x}) = \frac{1}{\sigma^2}\left(\frac{\int x_0\, q(x_0)\mathcal{N}(\tilde{x}; x_0, \sigma^2 I)\,dx_0}{\int q(x_0)\mathcal{N}(\tilde{x}; x_0, \sigma^2 I)\,dx_0} - \tilde{x}\right) = \frac{\mathbb{E}[x_0 \mid \tilde{x}] - \tilde{x}}{\sigma^2}.$$

Hence

$$\mathbb{E}[x_0 \mid \tilde{x}] = \tilde{x} + \sigma^2 \nabla_{\tilde{x}} \log p_\sigma(\tilde{x}).$$

5.2 Explicit denoiser–score mapping

Let $\hat{x}_0^*(\tilde{x}) = \mathbb{E}[x_0 \mid \tilde{x}]$, and $s_\sigma(\tilde{x}) = \nabla_{\tilde{x}} \log p_\sigma(\tilde{x})$. Then

$$\hat{x}_0^*(\tilde{x}) = \tilde{x} + \sigma^2 s_\sigma(\tilde{x}), \qquad s_\sigma(\tilde{x}) = \frac{\hat{x}_0^*(\tilde{x}) - \tilde{x}}{\sigma^2}.$$

5.3 Why predicting ϵ is often easier

Two precise senses:

- **Scale normalization:** $\epsilon \sim \mathcal{N}(0, I)$ has near unit scale across noise levels, while score magnitude scales roughly as $O(1/\sigma)$, creating large dynamic range as $\sigma \to 0$.
- **Gradient magnitude stability:** score targets can become very large at low noise, making optimization sensitive; ϵ-targets are more homoscedastic and typically better conditioned.

7 Problem 6: Unifying DDPM and Score Matching

6.1 Map ϵ-prediction to score estimate

Define noised marginal

$$q_i(x) = \int q(x_i = x \mid x_0)\, q(x_0)\, dx_0.$$

From

$$x_i = \sqrt{\bar{\alpha}_i} x_0 + \sqrt{1 - \bar{\alpha}_i}\epsilon,$$
$$\nabla_{x_i} \log q_i(x_i) = -\frac{1}{\sqrt{1 - \bar{\alpha}_i}} \mathbb{E}[\epsilon \mid x_i].$$

So an ϵ-predictor induces

$$\hat{s}_\theta(x_i, i) := -\frac{1}{\sqrt{1 - \bar{\alpha}_i}} \epsilon_\theta(x_i, i) \approx \nabla_{x_i} \log q_i(x_i).$$

6.2 Effective noise level

Rewrite

$$x_i = \sqrt{\bar{\alpha}_i} \left(x_0 + \sqrt{\frac{1 - \bar{\alpha}_i}{\bar{\alpha}_i}} \epsilon \right) = \sqrt{\bar{\alpha}_i}(x_0 + \sigma_i \epsilon),$$

thus

$$\sigma_i = \sqrt{\frac{1 - \bar{\alpha}_i}{\bar{\alpha}_i}}.$$

So $x_i / \sqrt{\bar{\alpha}_i}$ is exactly in additive-noise form $x_0 + \sigma_i \epsilon$.

6.3 Which parameterization is better conditioned in high dimensions?

Define conditioning as timestep-uniform target/gradient scale in the training loss.

- ϵ-prediction: generally stable, but converted x_0-error is amplified by $1/\sqrt{\bar{\alpha}_i}$ at low SNR.
- x_0-prediction: converted ϵ-error gets weight $\frac{1 - \bar{\alpha}_i}{\bar{\alpha}_i}$, which can explode at late noisy steps.
- v-prediction: mixes x_0, ϵ so scale is more uniform across SNR range; empirically and analytically best balanced.

Hence in high dimensions across wide SNR, v-parameterization is typically best conditioned, ϵ next, x_0 worst at high-noise steps.

8 Conclusion

All required derivations were given in assignment notation. The DDPM objective, posterior algebra, Tweedie identity, and NCSN/DDPM equivalence are internally consistent through the map

$$\epsilon_\theta \longleftrightarrow \text{score} \longleftrightarrow \text{denoiser}.$$

This directly supports both the analytical answers and the coding choices.